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Set P

**B.Sc. (E.C.S.) (Semester - III) (New) (NEP CBCS) Examination:
March/April – 2026
Data Structures using C++ (G06-DSC1-0301)**

Day & Date: Monday, 06-04-2026
Time: 09:00 AM To 10:30 AM

Max. Marks: 30

- Instructions:** 1) All questions are compulsory.
2) Draw neat diagrams and give equations wherever necessary.
3) Figures to the right indicate full marks.

Q.1 Choose correct alternatives.

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- 1) How many children does a node of binary tree have?
 - a) more than 2
 - b) any number of children
 - c) 0 or 1 or 2
 - d) 0 or 1
- 2) What is time complexity of linear search method to search 'n' element?
 - a) $O(1)$
 - b) $O(n)$
 - c) $O(\log n)$
 - d) $O(n^2)$
- 3) Quick sort method also called as ____
 - a) Partition Exchange sort
 - b) Selection Sort
 - c) Exchanging Sort
 - d) Merging Sort
- 4) Which of the following searching algorithm is fastest?
 - a) Binary search
 - b) Linear search
 - c) Jump search
 - d) All are equally fast
- 5) A normal queue, if implemented using an array of size MAX_SIZE, gets full when?
 - a) $\text{Rear} == \text{MAX_SIZE} - 1$
 - b) $\text{Front} == (\text{rear} + 1) \bmod \text{MAX_SIZE}$
 - c) $\text{Front} == \text{rear} + 1$
 - d) $\text{Rear} == \text{front}$
- 6) The data structure required to check whether an expression contains a balanced parenthesis or not?
 - a) Queue
 - b) Stack
 - c) Tree
 - d) Array

Q.2 Answer the following. (Any Three)

06

- a) Define Strictly binary tree.
- b) Evaluate the following POSTFIX expression using stack.
 $PQ + RS - * TU + /$ where $P = 2, Q = 1, R = 4, S = 5, T = 3, U = 7$
- c) Why queue is called FIFO data structure?
- d) List out applications of tree.
- e) What is Linear Circular linked list?

Q.3 Answer the following. (Any Two) 06

- a) Write CPP program to implement static queue.
- b) Perform radix sort on given data.
121, 23, 45, 232, 321, 36, 87, 63, 44, 101, 44
- c) Explain ADT for Array.

Q.4 Answer the following. (Any Two) 06

- a) Write a program to implement static STACK with following operations:
 - i) CREATE
 - ii) PUSH
 - iii) POP
 - iv) Display
- b) What is tree traversal? Explain Preorder and In-order traversing with example.
- c) Write CPP program to perform addition of two-dimensional arrays.

Q.5 Answer the following. (Any One) 06

- a) Write CPP Program to perform following operations on singly linear linked list.
 - i) Insert new node at beginning
 - ii) Remove last node
 - iii) Search node
 - iv) Display linked list
- b) What is Output Restricted Deque? Explain its following operations -
 - i) Insert Right
 - ii) Insert Left
 - iii) Remove
 - iv) Display